

Robinson's Transformation & Causal Forests

Step-by-Step: Confounding-Robust CATE Estimation

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The Problem: Why Naive Comparison Fails

What We Want — CATE

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$$

The **individual-level** treatment effect as a function of covariates x .

Naive Estimator is Biased

$$\hat{\tau}^{\text{naive}}(x) = \mathbb{E}[Y \mid X=x, W=1] - \mathbb{E}[Y \mid X=x, W=0]$$

Biased whenever W_i is correlated with X_i .

The Fix: Robinson's Transformation

Partial out nuisance functions *before* estimating $\tau(x)$.
Both bias sources vanish by construction.

Two Root Causes of Bias

1. Confounding Bias

Non-random treatment makes treated & control groups *structurally different*.

2. Regularization Bias

ML penalties affect treatment/control groups differently when $n_1 \neq n_0$.

Robinson's Transformation: The Partialling-Out Trick

Step 1 — Partially Linear Model

$$Y_i = \underbrace{\mu^{(0)}(X_i)}_{\text{nuisance baseline}} + \tau \cdot W_i + \epsilon_i$$

Step 2 — Conditional Expectation on X_i

$$\underbrace{\mathbb{E}[Y_i | X_i]}_{m(X_i)} = \mu^{(0)}(X_i) + \tau \cdot \underbrace{\mathbb{E}[W_i | X_i]}_{e(X_i)}$$

Step 3 — Subtract (the key trick)

$$\underbrace{Y_i - m(X_i)}_{\tilde{Y}_i} = \tau \cdot \underbrace{(W_i - e(X_i))}_{\tilde{W}_i} + \epsilon_i$$

Residual Intuition

- $\tilde{Y}_i = Y_i - \hat{m}(X_i)$

How surprising is this outcome given this person's features?

- $\tilde{W}_i = W_i - \hat{e}(X_i)$

How surprising is this treatment given this person's features?

From Global $\hat{\tau}$ to Local $\hat{\tau}(x)$

Global OLS Estimator:

$$\hat{\tau} = \frac{\sum_i \tilde{Y}_i \tilde{W}_i}{\sum_i \tilde{W}_i^2}$$

- Single global treatment effect
- Units with $|\tilde{W}_i| \approx 0$ contribute little
- **Key insight:** Large treatment surprise drives estimation

Limitation

Ignores local heterogeneity in treatment effects

Local Robinson Estimator:

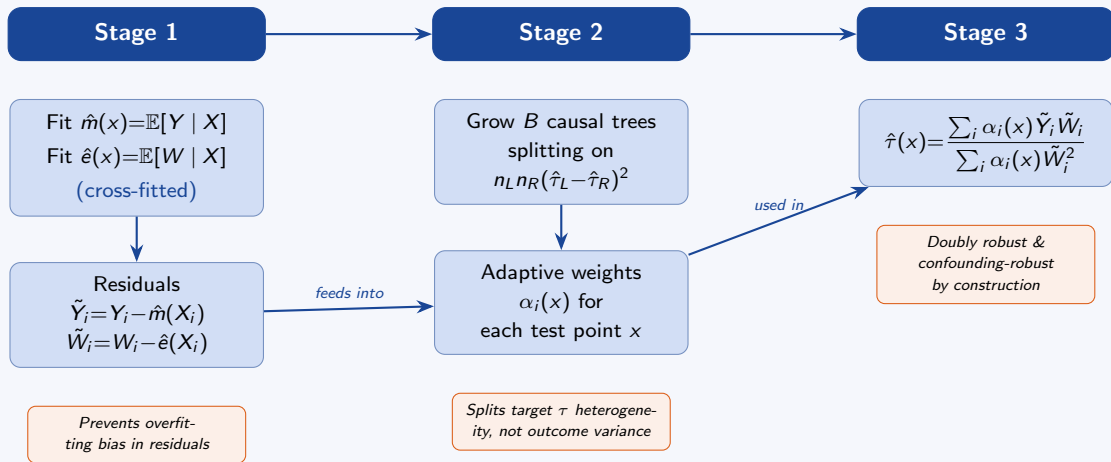
$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x) \tilde{Y}_i \tilde{W}_i}{\sum_i \alpha_i(x) \tilde{W}_i^2}$$

- Forest weights $\alpha_i(x)$ create adaptive neighborhoods and measures proximity of X_i to $x \Rightarrow$ confounding-robust

Forest Weights

$$\alpha_i(x) = \frac{1}{B} \sum_{b=1}^B \frac{\mathbf{1}[X_i, x \in \text{leaf}_b]}{|\text{leaf}_b(x)|}$$

The Full Three-Stage Causal Forest Pipeline



Why Robinson's Transformation Works

1. **Partialling out** eliminates unknown baseline $\mu^{(0)}(x)$ exactly
2. Residuals encode *only* variation unexplained by X
3. Units with large $|\tilde{W}_i|$ drive estimation by design
4. **Double robustness** tolerates misspecification

Why Causal Forests Extend This Properly

- Weights $\alpha_i(x)$ make estimation genuinely **local**
- Splits target τ heterogeneity — not spurious outcome patterns
- Robinson's form preserved locally $\Rightarrow$ **confounding-robust**